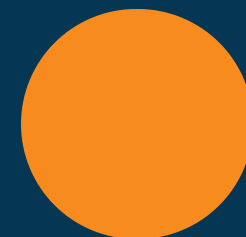




# The Next Generation of ReShare

Architecture, adaptable workflows, interoperability, and migration

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# From CrossLink broker to CrossLink platform

| WOLFcon 2025                   | WOLFcon 2026                    |
|--------------------------------|---------------------------------|
| Standards-based ILL broker     | Complete ILL workflow platform  |
| Connect external peers         | Native borrowing and lending    |
| Route and observe transactions | Model and operate the lifecycle |

New: Patron Requests API · NCIP · explicit state model · live UI events · scheduling

## What ReShare proved

- Community-owned resource sharing works
- ILS-neutral workflows can operate in production
- Practitioners can shape the service
- Open standards connect diverse systems

## What needed to change

- Substantial application and persistence framework overhead
- Tenant topology multiplied operational work
- Kafka added infrastructure and failure windows
- Workflow behavior was spread across the application
- Workflow changes required tracing, releases, and broad testing

**We are replacing an implementation, not the product.**

## Preserve

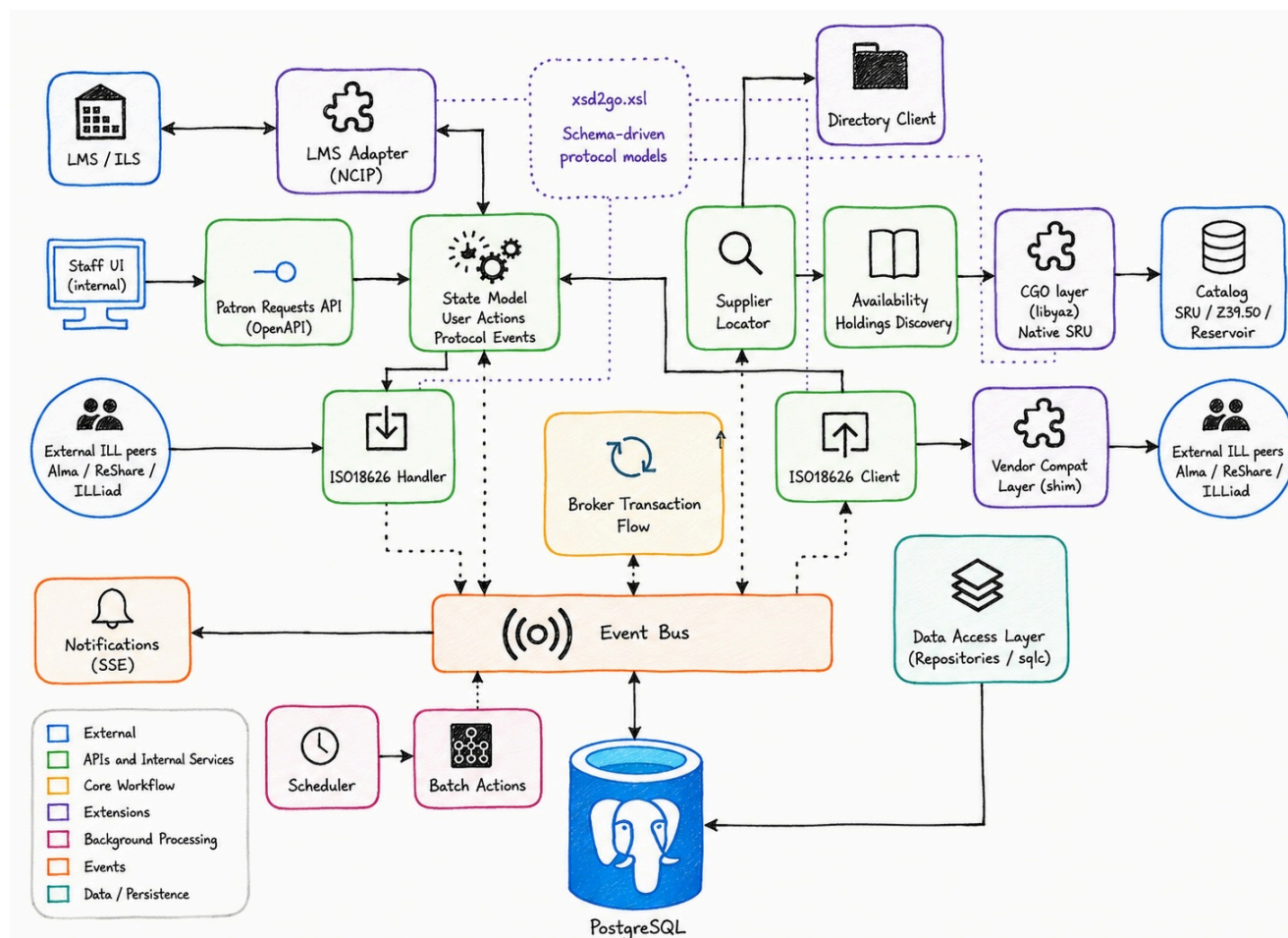
- ILS/LSP neutrality
- First-class ISO 18626 and NCIP
- Borrowing and lending workflows
- Consortial routing and supplier selection
- Multi-tenant operation

## Improve

- Smaller, more reliable operational footprint
- Explicit and adaptable workflows
- Observable and auditable requests
- One shared deployment for virtual tenants
- Support for native direct borrowing across the consortium
- A richer, OpenAPI-based Directory for networks, tiers, policies, and integrations (NCIP, Z39.50)

# A platform built around the transaction

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Open protocols at the edges · explicit workflow at the center · durable work underneath

# A small core supports a complete platform

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## Lightweight core runtime

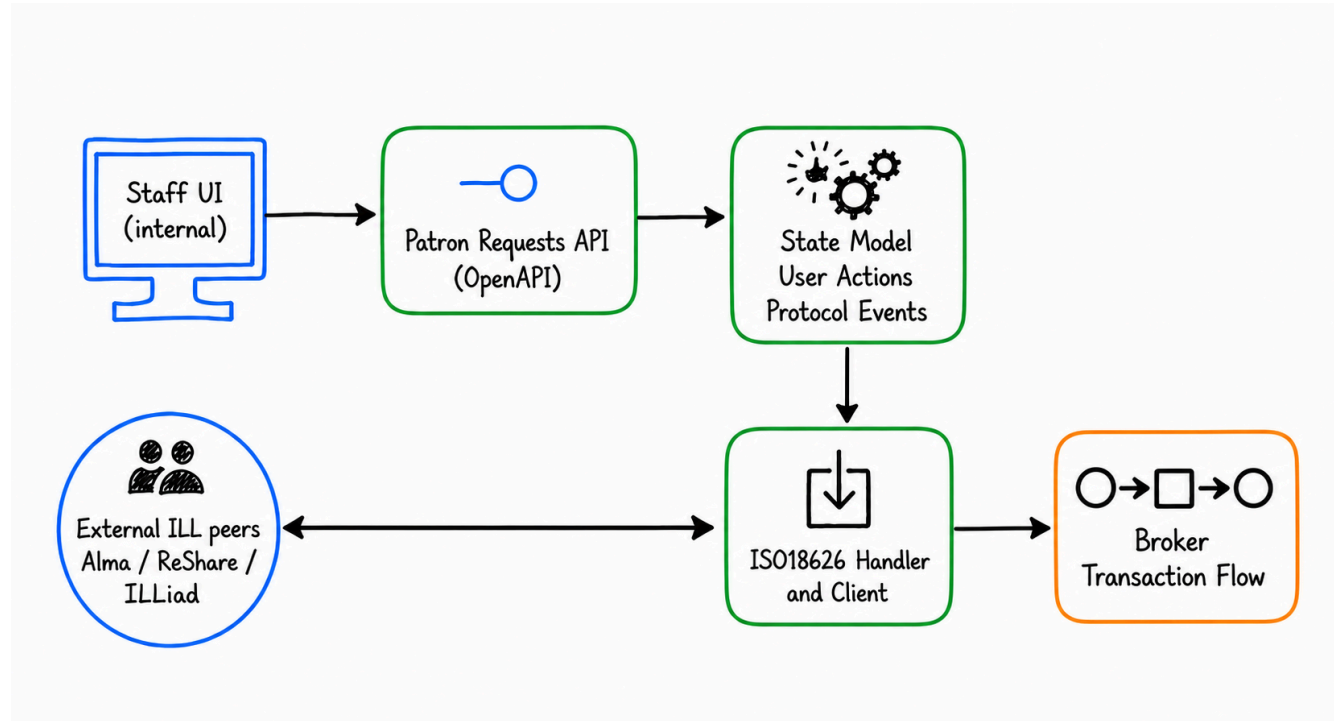
- One Go service around durable state
- PostgreSQL as the only infrastructure dependency
- No Kafka cluster for core messaging
- Custom event bus implementation
- Stateless Kubernetes/Helm deployment and health endpoints
- Packaged database migrations and rolling updates

## Main features built on the shared core

- Workflow execution
- Scheduler and batch actions
- Request aging and retries
- Notifications and email templates
- Pull slips and document delivery
- Live UI updates through SSE

# One request, two complementary views

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## Patron Request

- The lifecycle practitioners operate
- Borrowing or lending perspective
- Current state and available actions
- Items, notifications, and attention flags
- Search, filtering, and batch work

## ILL Transaction

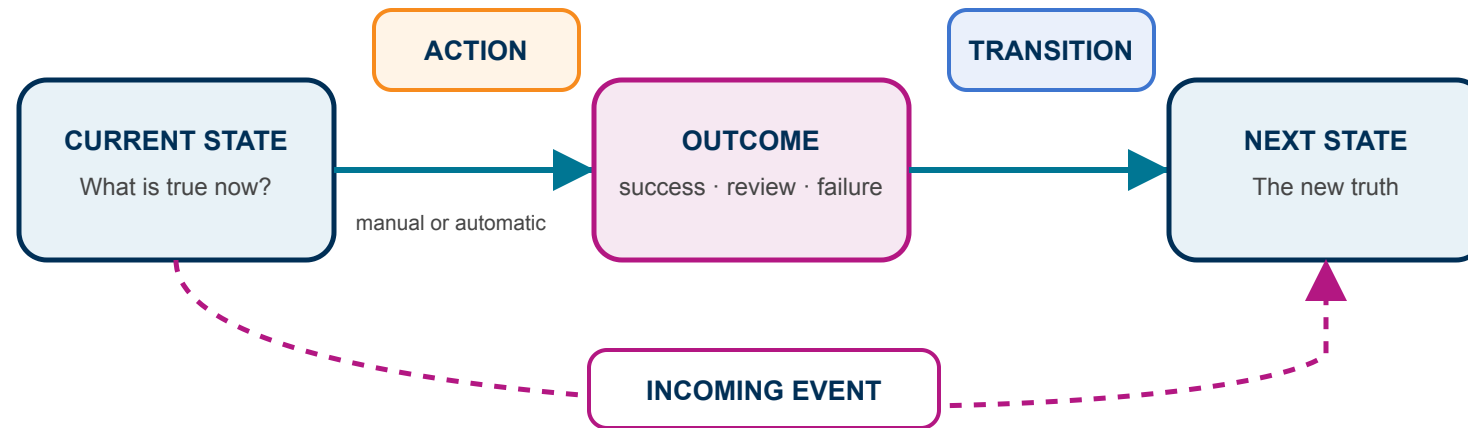
- Supplier and rota decisions
- ISO 18626 message exchange
- Adapter and protocol activity
- Complete durable event history

**A hyperlinked API connects the request, transaction, protocol activity, and event history—from intent to outcome.**

# Workflow is now a YAML model

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## A small grammar for a complete workflow

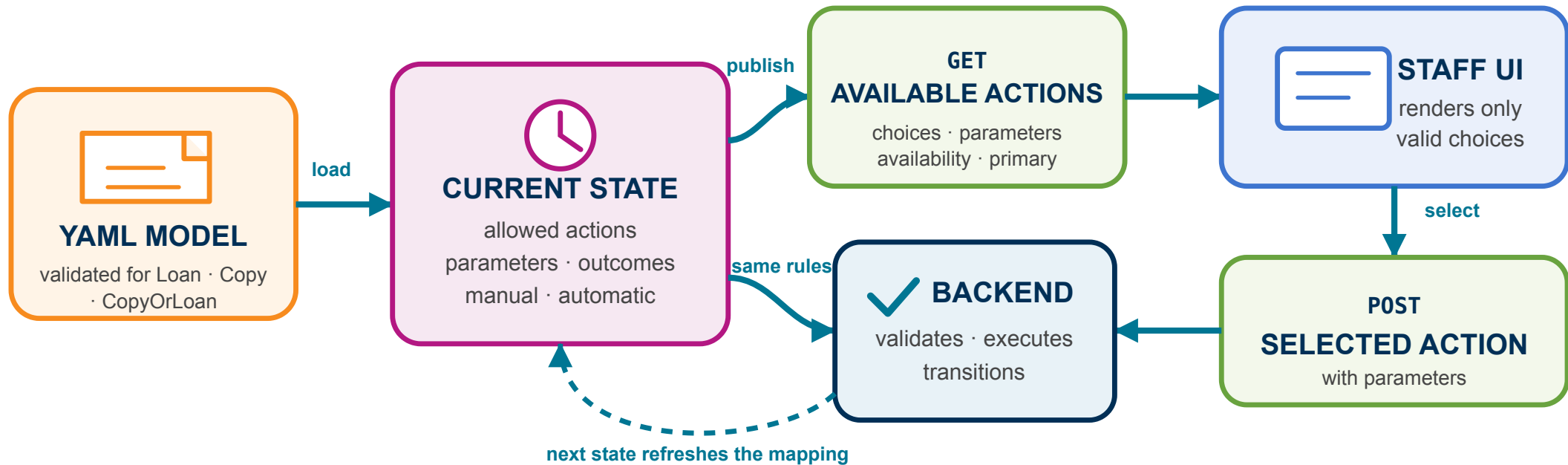


External messages use the same permitted transitions

```
- name: NEW
  side: REQUESTER
  initial: true
  primaryAction: validate-patron
  actions:
    - name: validate-patron
      trigger: auto
      transitions: {success: VALIDATED, review: INVALID_PATRON}
```

# One workflow model drives the API and UI

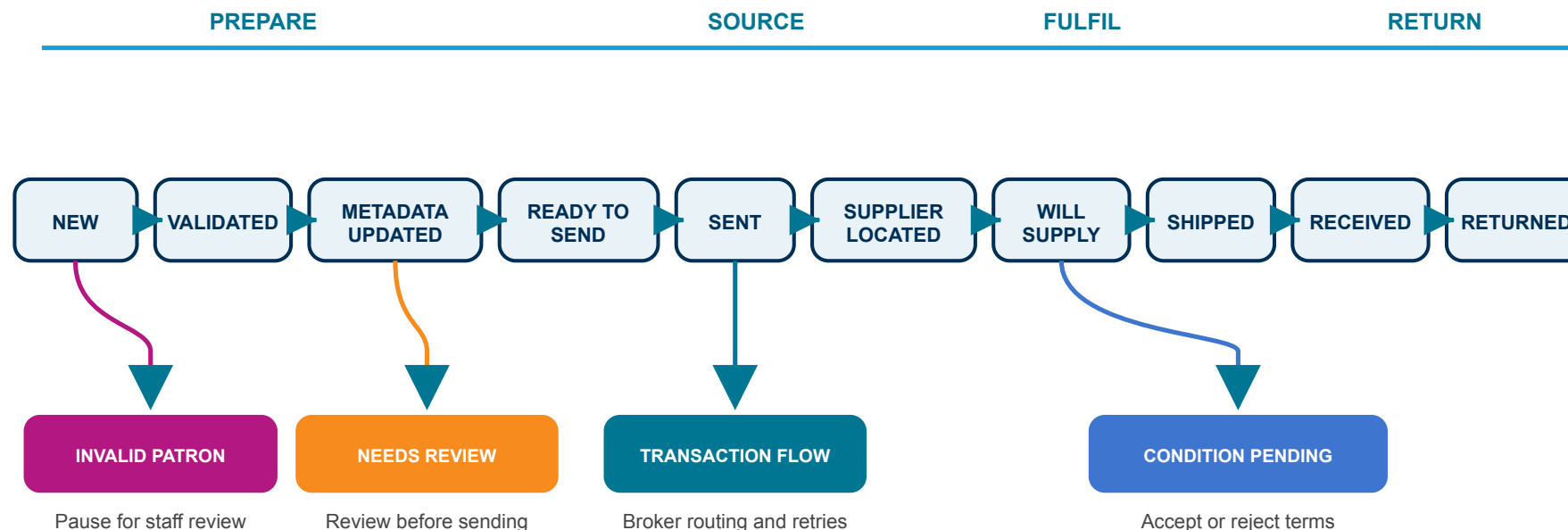
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Accepted actions become durable database events: recoverable after failure, observable through the API, and safely distributed across broker instances.

# A borrowing request, step by step

WOLFcon 2026



The happy path is simple. The model makes exceptions explicit.

**The happy path and every exception are explicit, inspectable, and testable.**

## Shape the workflow

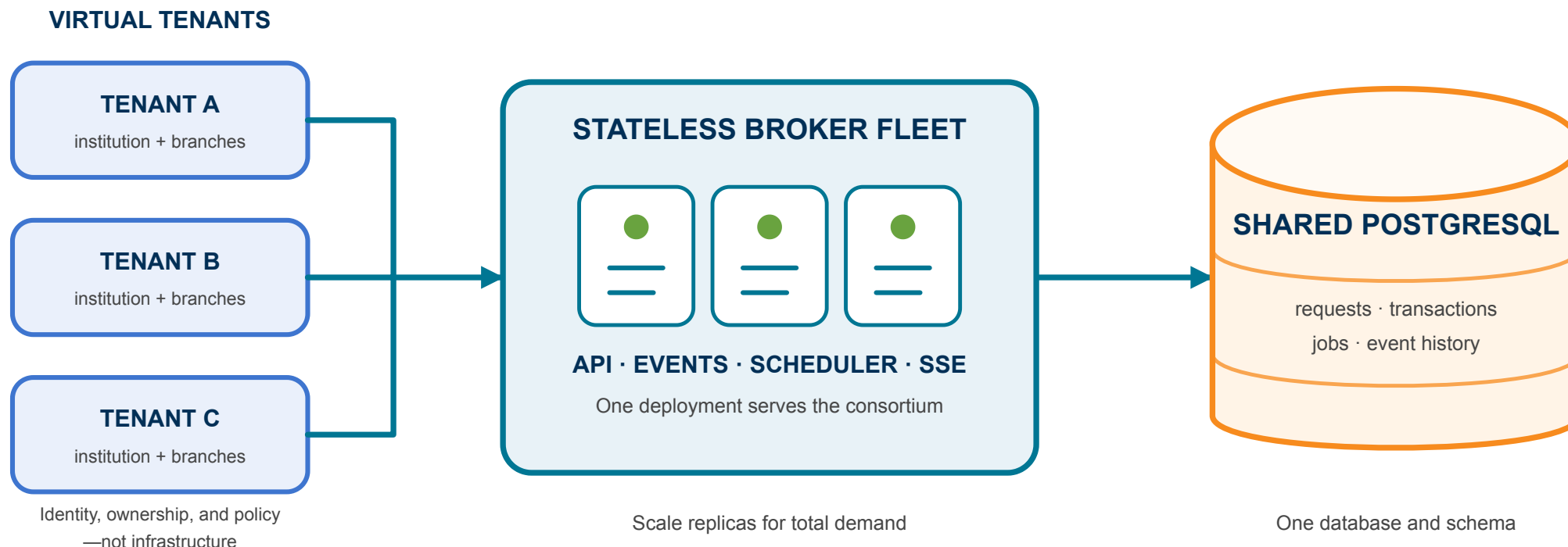
- Choose supported actions per state
- Make actions manual or automatic
- Map outcomes and incoming messages
- Apply behavior by service type
- Mark primary, editable, terminal, and attention states

## Enforce the contract

- Publish supported capabilities
- Reject unknown actions and events
- Prevent requester/supplier crossovers
- Validate transitions and closing paths
- Check service-type consistency

# One deployment, many virtual tenants

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**The new architecture enables native DCB support in ReShare**

Union catalog → live availability → internal fulfillment → local ILS

**A shared deployment is an option, not a constraint**  
Independent local brokers can still form a peer-to-peer ILL network

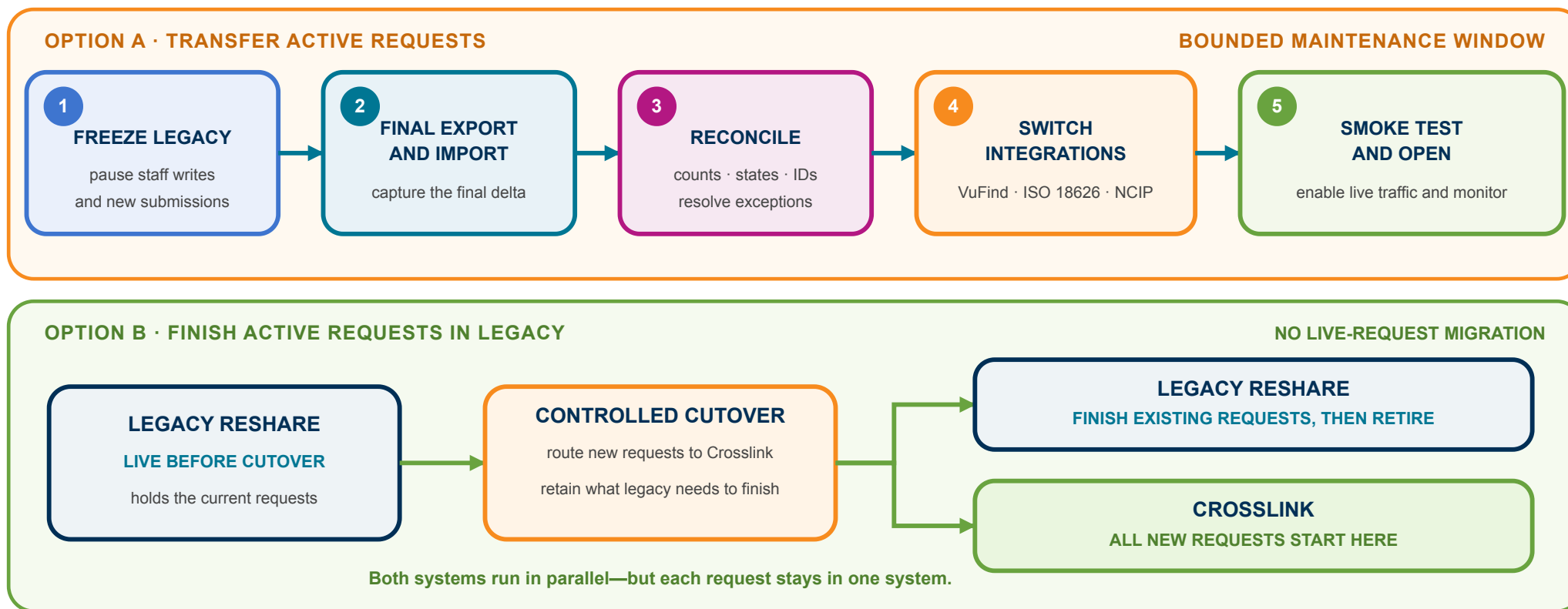
| Measure                      | CrossLink | mod-rs      | Difference          |
|------------------------------|-----------|-------------|---------------------|
| Container image              | 35.6 MB   | 246.0 MB    | <b>6.9× smaller</b> |
| Slowest cold start—3 runs    | 0.667 s   | 37.006 s    | <b>55× faster</b>   |
| App working set (RSS)        | 19.7 MiB  | 1,509.2 MiB | <b>77× lower</b>    |
| Core runtime pieces          | 2         | 5           | <b>3 fewer</b>      |
| Test-to-production LOC ratio | 1.71:1    | 0.16:1      | <b>10.9× higher</b> |

Native ARM and a 2 GiB app limit. Legacy app tier includes Kafka + ZooKeeper; PostgreSQL and Okapi excluded.

**A smaller operational footprint means simpler deployment, recovery, scaling, and upgrades.**

# Adoption with a migration plan

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Rehearse with real data, prove readiness, and choose the cutover path that fits each consortium.

# A platform that evolves with the service

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- Open at the edges
- Explicit at the center
- Small enough to operate
- Flexible enough to evolve
- A migration path that protects continuity

**The next generation of ReShare makes the community's ILL workflow a first-class part of the platform.**